Symmetric Smoothing Filters From Global Consistency Constraints

Gautam Pai

13th April 2015

1. Introduction and Previous Work
2. Global Consistency Constraints
3. A Regularization Framework: The Consistency Filter
4. Experiments and Results

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Introduction and Previous Work

- Milanfar, Peyman. "A tour of modern image filtering." IEEE Signal Processing Magazine 2 (2011).
- Milanfar, Peyman. "Symmetrizing smoothing filters." SIAM Journal on Imaging Sciences 6.1 (2013): 263-284.

Problem Formulation

The Noise Model

$$y_i = z_i + n_i$$

y_i: noisy signal value at pixel i
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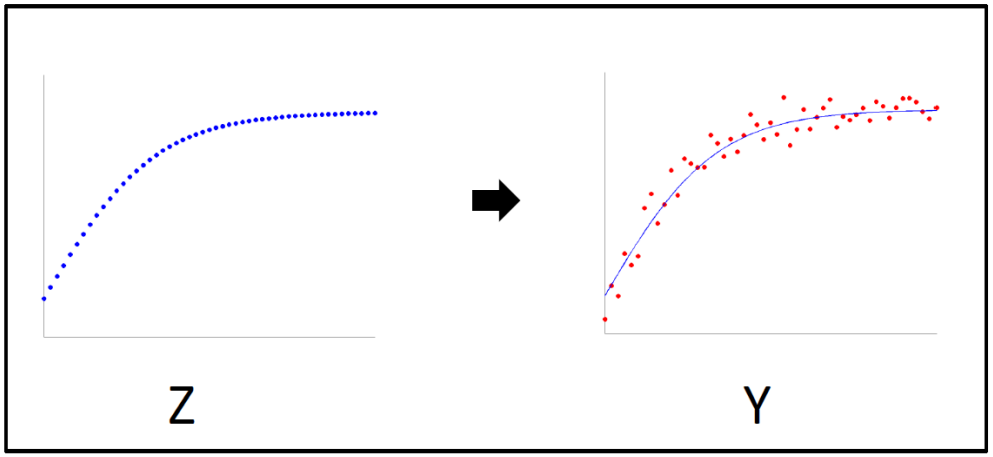
$$\therefore \; Y = Z + N, \; where \; N \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I})$$

Problem Formulation: Visualizing 1D

PERTURBATION MODEL



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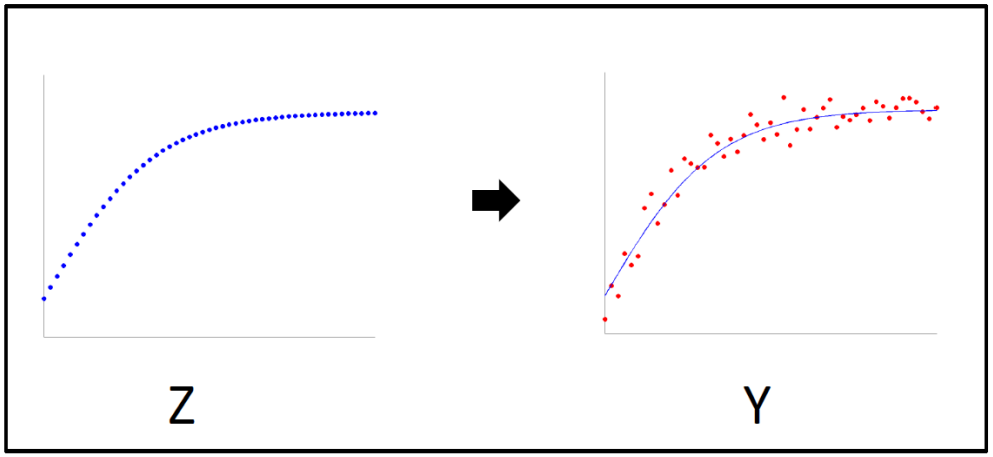
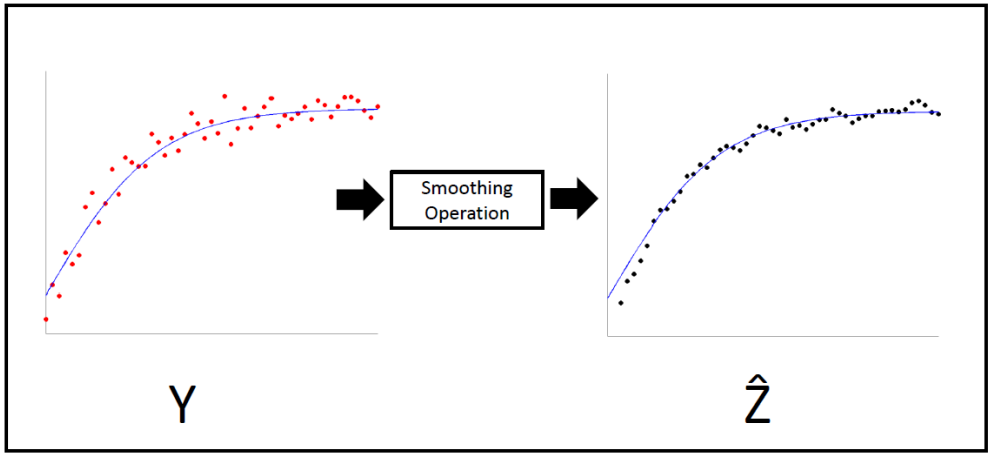


OBJECTIVE



A Simple Filtering Scheme

Multidimensional Filtering

$$\hat{z}_i = \sum_j W_{ij}\, y_j$$

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- functional form of the weights W_{ij} characterize the filter
- for the standard box filter:

$$\begin{aligned} W_{ij} &= \frac{1}{P_s^2} : \quad if\,|x_i - x_j| < P_s \\ &= 0 : \quad otherwise \end{aligned}$$

where, x_i denotes spatial position of pixel i

A Generic Framework for Smoothing

The Weighted Least Squares Problem

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- **Non-Local Means**

$$K(i,j) = e^{\frac{-||\mathbf{P_i} - \mathbf{P_j}||^2}{2\sigma^2}}$$

similarity based only on **photometric distance of patches-$\mathbf{P_i}$, $\mathbf{P_j}$**

A Unified Framework

if,

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stacking the estimates,

Multi-Dimensional Filtering

$$\widehat{Z} = W\ Y$$

The Graph Perspective

For a given similarity kernel $K(i, j)$,

$$K = \begin{bmatrix} K(1,1) & K(1,2) & \dots & K(1,MN) \\ K(2,1) & K(2,2) & \dots & K(2,MN) \\ . & . & . & . \\ . & . & . & . \\ K(MN,1) & K(MN,2) & \dots & K(MN,MN) \end{bmatrix}$$

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$$D = \begin{bmatrix} D(1,1) & 0 & \dots & 0 \\ 0 & D(2,2) & \dots & 0 \\ . & . & . & . \\ . & . & . & . \\ 0 & 0 & \dots & D(MN,MN) \end{bmatrix}$$

where, $D(i,i) = \sum_{j=1}^{j=MN} K(i,j)$

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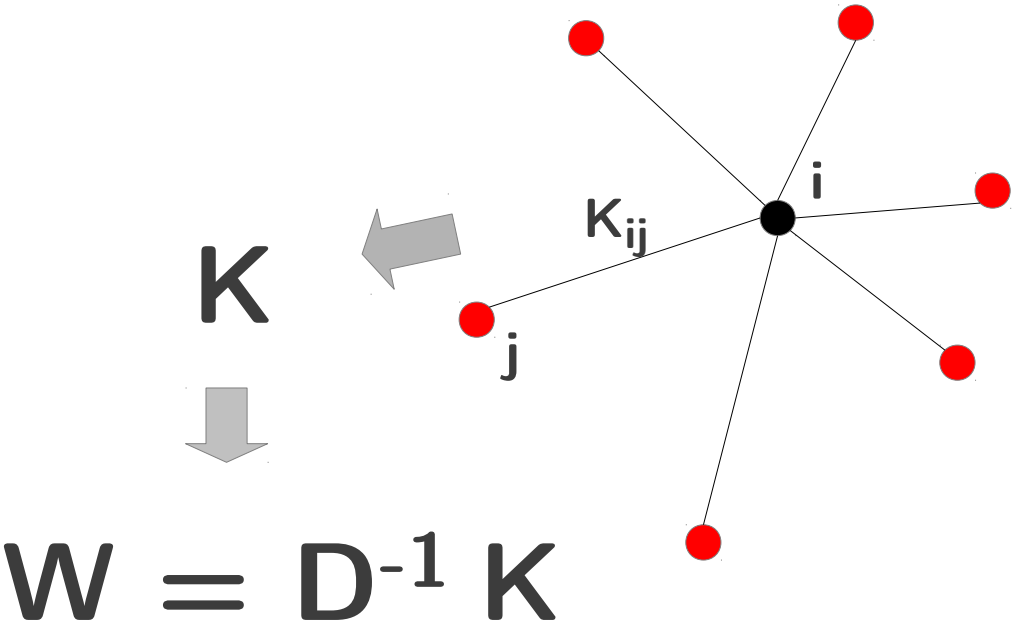

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- W encapsulates the geometry of the image in a manner defined by the kernel $K(i, j)$

Properties of W we would like to have

- **Symmetry** - In every pixel pair, the two pixels affect each others output equally

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- Give Better Results!

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- **W** encodes **global** information from **local** relationships
- However, these relationships are not necessarily maintained post the filtering operation
- if filter **W** is some oracle encoding all local relationships and $\widehat{Z}$ is the filtered output then we should expect:

$$\widehat{Z} = \mathbf{W}\widehat{Z}$$

- The pairwise geometric relationships in $\widehat{Z}$ are **consistent**

Global Consistency Constraints

- gives an interpretation of the idea that signal and noise have been completely separated

$$\widehat{Z} = \mathbf{W}\widehat{Z} = \mathbf{W}^2\widehat{Z} = \mathbf{W}^3\widehat{Z} = \ldots$$

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A De-Noising Strategy

$$\widehat{Z} = \underset{Z}{arg\,min}\ ||Z - Y||^2 \ \ subject\ to\ \ Z = \mathbf{W}Z$$

Connections to Spectral Graph Theory

Looking closely at our consistency constraint again:

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- matrix $L = I - \mathbf{W} = I - D^{-1}K$ is one of the many versions of the Graph Laplacian
- therefore global consistency requires that the estimate lie in the null space of the Graph-Laplacian L

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- The multiplicity k of the eigenvalue 0 of L equals the number of connected components $\{A_1, ..., A_k\}$ in the graph.

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effectively, global consistency requires:

$$\widehat{Z} = \alpha_1 \, \mathbf{1_{A_1}} + \alpha_2 \, \mathbf{1_{A_2}} + \alpha_3 \, \mathbf{1_{A_3}} \cdots + \alpha_k \, \mathbf{1_{A_k}}$$

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where

$$\alpha_i = \frac{Y^T \mathbf{1_{A_i}}}{|\mathbf{1_{A_i}}|} \;\; \forall \, i = 1, 2..k$$

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- Solution: penalize deviation from consistency in a regularization framework

$$\widehat{Z} = \underset{Z}{arg\,min} \; ||Z - Y||^2 + \lambda \, ||LZ||^2$$

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- Improve regularization using Weight Matrix Λ to account for different pixel connectivities

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- $\mathbf{L}$ is the un-normalized laplacian matrix: $\mathbf{L} = D - K$

Regularization

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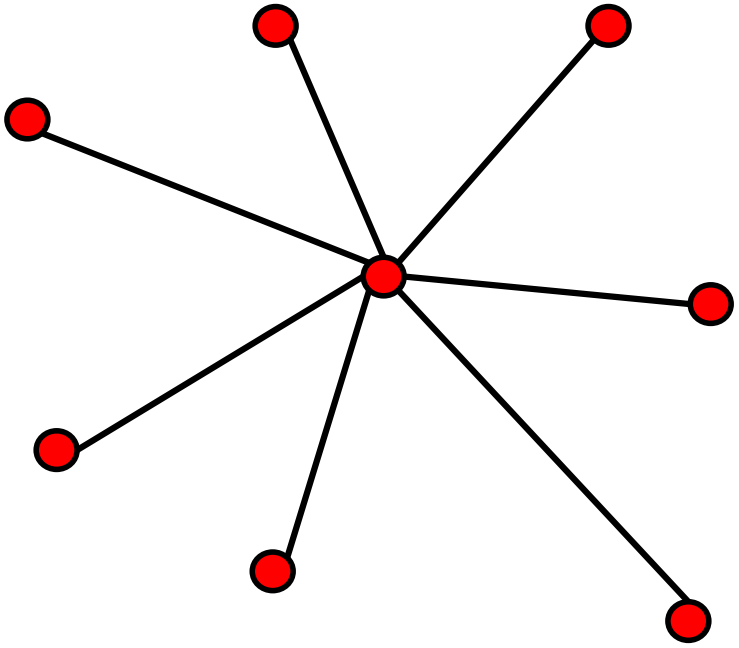
Y



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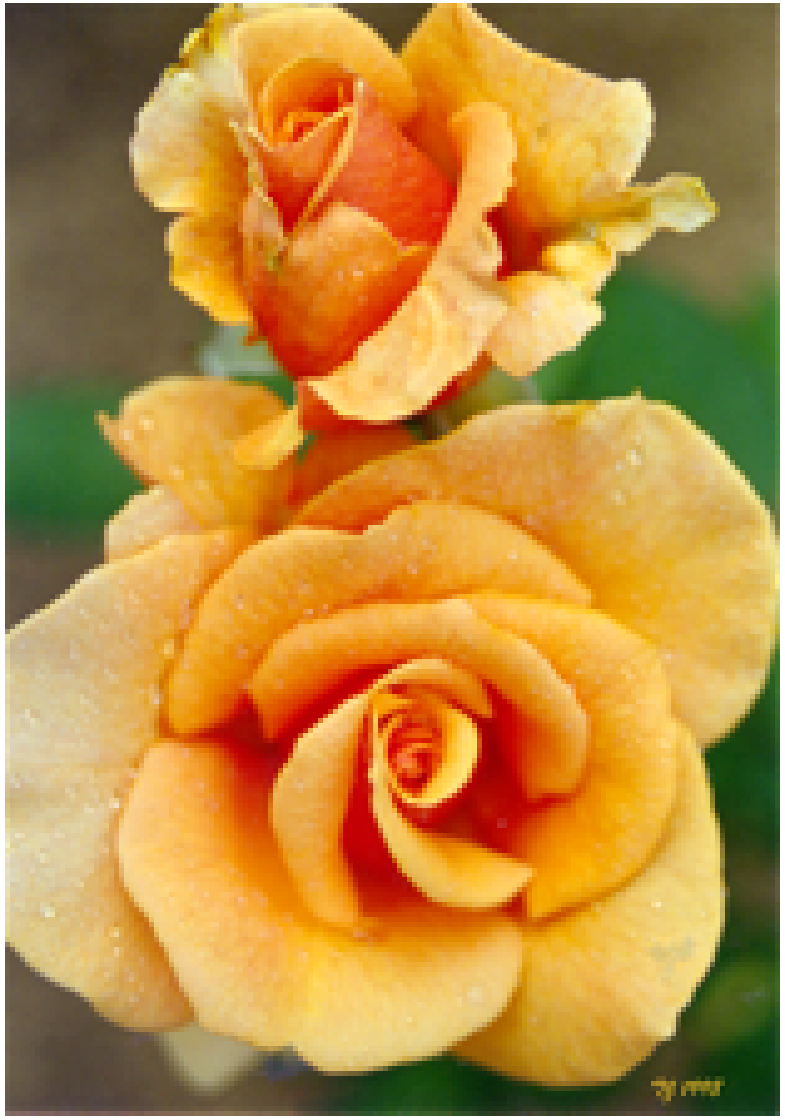
$$\mathbf{V_k}$$

$$\mathbf{Y_k} = \mathbf{V_k}\mathbf{V_k^T}\,\mathbf{Y} = \sum_{\mathbf{i=1}}^{\mathbf{k}} \left(\mathbf{V_i^T}\,\mathbf{Y} \right)\ \mathbf{V_i}$$



Original Image

k=1

k=5

k=10

k=20

k=40

k=60

k=100

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Setting the gradient to zero we get the system of equations:

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$$\therefore\ C^{-1}\,\mathbf{1} = \mathbf{1} \implies C\mathbf{1} = \mathbf{1}$$

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- C is a shrinking smoother: $||CY|| \leq ||Y||$

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\widehat{Z} &= CY \\
\widehat{Z} &= \Big[\sum_{i=1}^{i=MN} \frac{1}{1+\lambda\,\sigma_i^2}\, \mathbf{v_i}\mathbf{v_i}^T \Big] Y \\
\widehat{Z} &= \sum_{i=1}^{i=MN} \left[\frac{\mathbf{v_i}^T Y}{1+\lambda\,\sigma_i^2} \right] \mathbf{v_i}
\end{aligned}
$$

Properties of the Consistency Filter

$$
\begin{aligned}
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\widehat{Z} &= \sum_{i=1}^{i=MN} \left[\frac{\mathbf{v_i}^T Y}{1+\lambda\,\sigma_i^2} \right] \mathbf{v_i}
\end{aligned}
$$

Eigenvectors of the Graph Laplacian define the action of the Filter. The Consistency filter performs a shrinkage in this data-adaptive basis

Properties of the Consistency Filter

- The consistency filter can be expressed as the linear combination of rank-1 orthonormal matrices- **components**

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- λ gives the proportion of shrinkage. λ can also be interpreted as an index of diffusion

$$\lim_{\lambda \to 0} C = \mathbf{I}$$

$$\lim_{\lambda \to \infty} C = \sum_{i=1}^{i=k} \frac{1}{|A_i|} \mathbf{1}_{\mathbf{A_i}} \mathbf{1}_{\mathbf{A_i}}{}^T$$

Properties of the Consistency Filter

- The consistency filter can be expressed as the linear combination of rank-1 orthonormal matrices- **components**
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- Eigenvectors corresponding to smaller eigenvalues of Graph Laplacian **L** have the most influence in the output

1 Introduction and Previous Work

2 Global Consistency Constraints

3 A Regularization Framework: The Consistency Filter

4 Experiments and Results

We compare the performance of our filter with two other *symmetrizing* approaches:

- Sinkhorn Iterations for Symmetrizing Smoothing filters - **S**

$$\mathbf{S} = \mathbf{R}.\ \mathbf{W}.\ \mathbf{C}$$

where **R** and **C** are diagonal matrices and **S** is symmetric doubly stochastic

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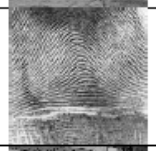
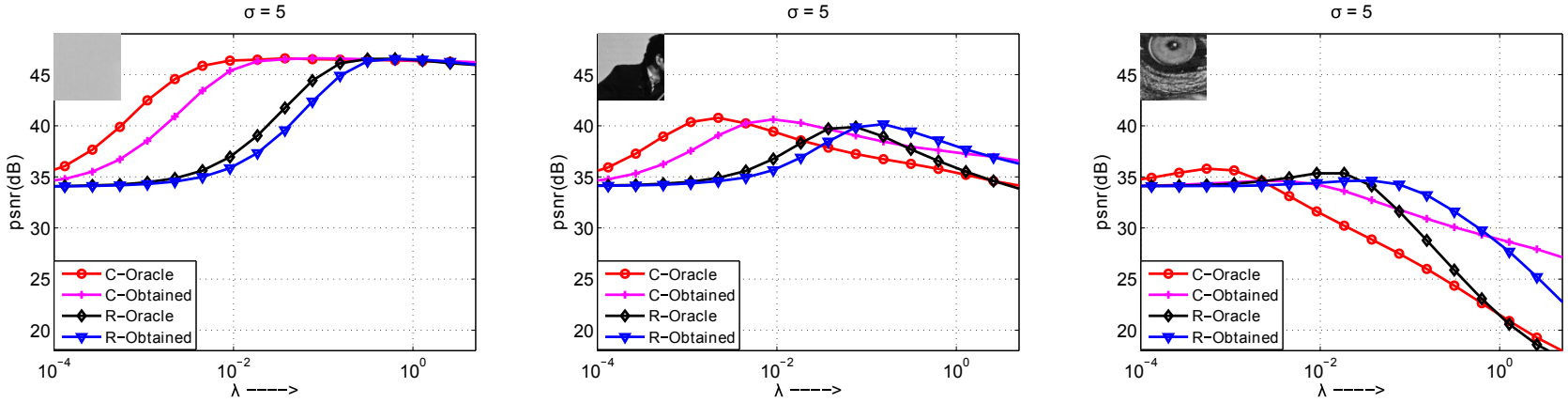
- Standard Laplacian Regularization

$$\begin{aligned} \widehat{Z} &= \underset{Z}{argmin}\ ||Z - Y||^2 + \lambda\ Z^T \mathbf{L} Z \\ \widehat{Z} &= \mathbf{R}\ Y \\ \mathbf{R} &= (\mathbf{I} + \lambda \mathbf{L})^{-1} \end{aligned}$$

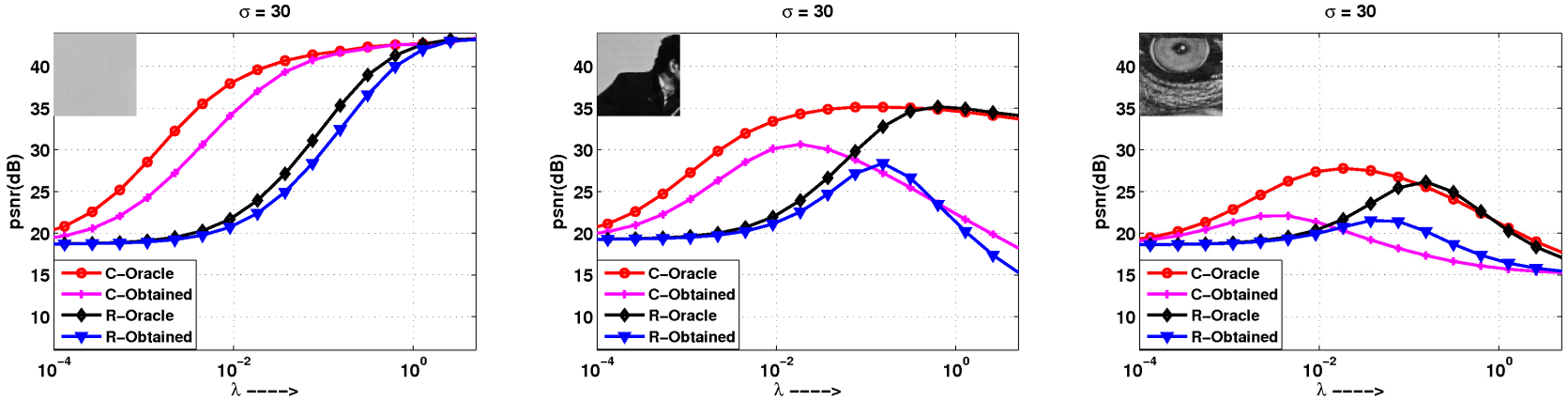
Comparison With Sinkhorn Iterations-**S**

Image	Noise Level	$\mathbf{W} = \mathbf{D}^{-1}\mathbf{K}$	$\mathbf{S}(\mathbf{W})$	**Our C filter**	**S(W)** Diff.	Our Diff.
	$\sigma = 20$	28.69	28.48	30.44	-0.21	1.75
	$\sigma = 40$	23.24	22.74	26.82	-0.50	3.58
	$\sigma = 20$	27.27	27.11	28.61	-0.16	1.34
	$\sigma = 40$	22.42	21.97	25.73	-0.45	3.31
	$\sigma = 20$	24.78	24.89	26.18	0.11	1.40
	$\sigma = 40$	22.16	21.96	23.08	-0.20	0.92
	$\sigma = 20$	24.89	25.29	27.35	0.40	2.46
	$\sigma = 40$	21.83	21.73	22.39	-0.10	0.56
	$\sigma = 20$	28.38	27.91	33.12	-0.47	4.74
	$\sigma = 40$	22.68	22.13	29.55	-0.55	6.87
	$\sigma = 20$	28.26	27.98	30.67	-0.28	2.41
	$\sigma = 40$	22.93	22.41	27.33	-0.52	4.40
	$\sigma = 20$	28.20	27.93	30.53	-0.27	2.33
	$\sigma = 40$	22.95	22.43	27.23	-0.52	4.28

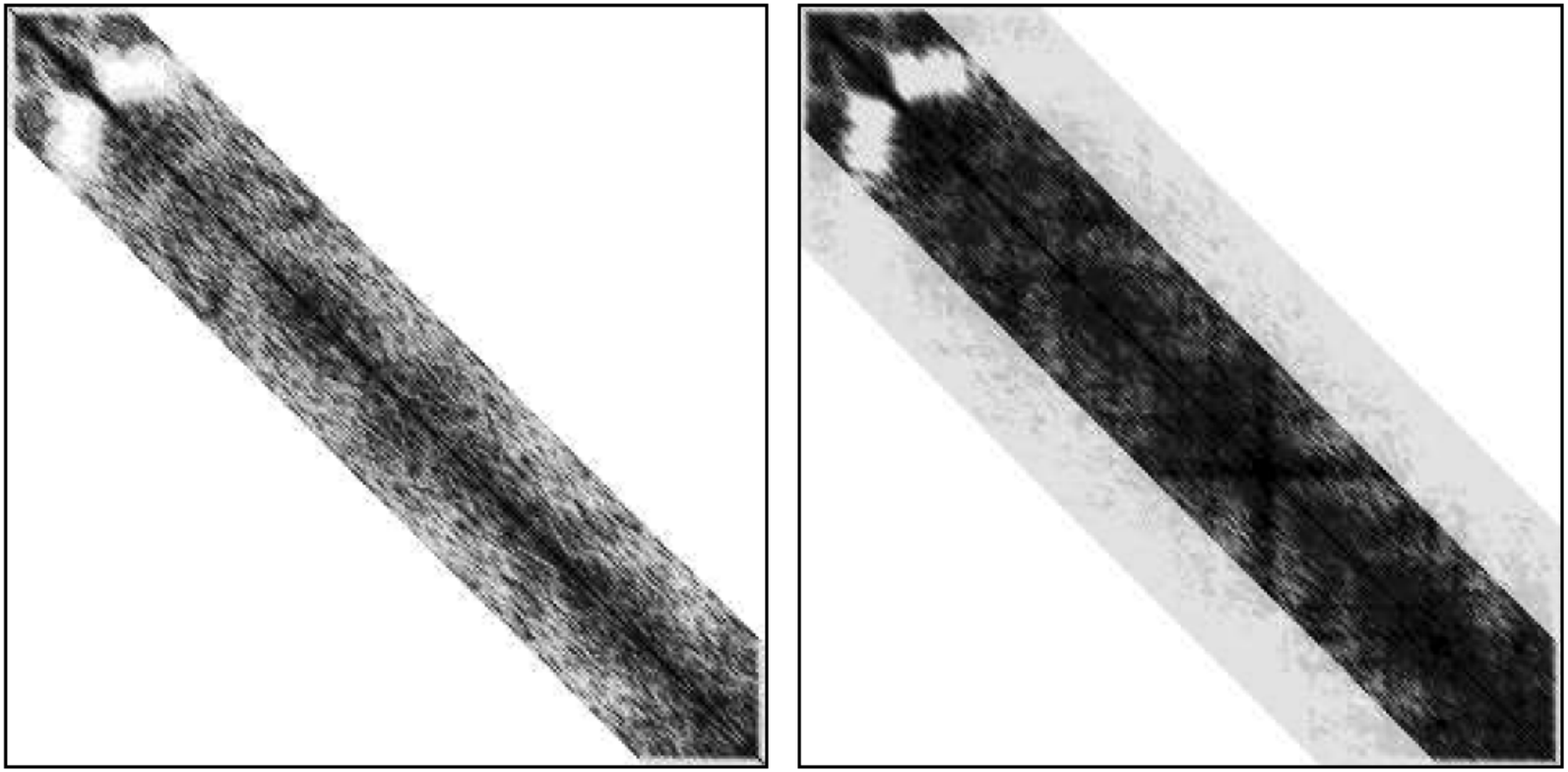
Comparison with Regular Laplacian Regularization **R**



(a) $\sigma = 10$



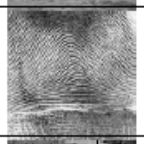
(b) $\sigma = 30$



(a) $\mathbf{L}$ (b) $\mathbf{L}^T\mathbf{L}$

Matrix visualizations of $\mathbf{L}$ and $\mathbf{L}^T\mathbf{L}$ for Lena image

Final Results Comparison

Image	Noise Level	$\mathbf{W} = \mathbf{D}^{-1}\mathbf{K}$	$\mathbf{S}(\mathbf{W})$ [7]	$\mathbf{R}$ filter $= (\mathbf{I}+\lambda\mathbf{L})^{-1}$	K-SVD [10]	BM3D [11]	Our $\mathbf{C}$ filter $= (\mathbf{I}+\lambda\mathbf{L}^T\mathbf{L})^{-1}$
	$\sigma = 20$	28.69	28.48	28.84	30.33	30.92	30.44
	$\sigma = 40$	23.24	22.74	26.00	26.37	27.43	26.82
	$\sigma = 20$	27.27	27.11	28.26	29.41	29.66	28.61
	$\sigma = 40$	22.42	21.97	24.61	25.56	26.25	25.73
	$\sigma = 20$	24.78	24.89	25.54	26.53	26.79	26.18
	$\sigma = 40$	22.16	21.96	21.96	22.77	23.23	23.08
	$\sigma = 20$	24.89	25.29	26.24	26.97	27.56	27.35
	$\sigma = 40$	21.83	21.73	21.37	21.96	22.70	22.39
	$\sigma = 20$	28.38	27.91	31.39	33.03	33.64	33.12
	$\sigma = 40$	22.68	22.13	27.88	29.20	30.66	29.55
	$\sigma = 20$	28.26	27.98	29.09	30.94	31.43	30.67
	$\sigma = 40$	22.93	22.41	26.10	27.01	27.93	27.33
	$\sigma = 20$	28.20	27.93	29.84	30.96	31.44	30.53
	$\sigma = 40$	22.95	22.43	26.17	27.34	28.18	27.23

Questions ?

Thank You!